

Algebra II with Trigonometry

Chapter 7.2

Study guide

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[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Certainly! Below is a comprehensive, self-contained study guide and response based on the provided PDF content for Chapter 7.2: Addition and Subtraction Formulas.

1. Short Summary

This document is a lesson outline for Algebra 2 Trigonometry/Honors, focusing on **addition and subtraction formulas** for sine, cosine, and tangent. It teaches students to:

- Apply these formulas to evaluate and simplify trigonometric expressions,
- Prove trigonometric identities using these formulas,
- Manipulate expressions involving inverse trig functions,
- Combine terms of the form $A \sin x + B \cos x$ into a single trig function,
- And solve related problems.

2. Core Definitions, Theorems, and Formulas

Addition and Subtraction Formulas

Sine:

$$\sin(s + t) = \sin s \cos t + \cos s \sin t$$

$$\sin(s - t) = \sin s \cos t - \cos s \sin t$$

Cosine:

$$\cos(s + t) = \cos s \cos t - \sin s \sin t$$

$$\cos(s - t) = \cos s \cos t + \sin s \sin t$$

Tangent:

$$\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$$

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

Expressions of the Form $A \sin x + B \cos x$:

Can be rewritten as a single sine function:

$$A \sin x + B \cos x = k \sin(x + \phi)$$

where:

$$k = \sqrt{A^2 + B^2}$$

$$\sin \phi = \frac{B}{\sqrt{A^2 + B^2}}$$

$$\cos \phi = \frac{A}{\sqrt{A^2 + B^2}}$$

Inverse Trigonometric Expressions

An expression like $\cos^{-1} x$ represents an **angle**.

3. Worked Study Guide with Step-by-Step Explanations

A. How to Use Addition and Subtraction Formulas

1. Evaluating Expressions

- When you see something like $\sin(75^\circ)$, write it as $\sin(30^\circ + 45^\circ)$ and apply the addition formula:

$$\sin(75^\circ) = \sin(30^\circ + 45^\circ) = \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$$

2. Proving or Simplifying Identities

- If given $\cos(x - y)$, expand to $\cos x \cos y + \sin x \sin y$.
- Rearranging and factoring may help verify a given identity.

3. Combining Expressions into a Single Trig Function

- For $3 \sin x + 4 \cos x$:
- $k = \sqrt{3^2 + 4^2} = 5$
- $\sin \phi = \frac{4}{5}$, $\cos \phi = \frac{3}{5}$
- So, $3 \sin x + 4 \cos x = 5 \sin(x + \phi)$, where $\sin \phi = 4/5$
- Find ϕ using $\arcsin(4/5)$ if needed.

4. Working with Inverse Trig Expressions

- $\sin(\cos^{-1} x)$ can be written using a right triangle:
- Let $\cos^{-1} x = \theta$ so $\cos \theta = x$.
- Draw a right triangle with adjacent x , hypotenuse 1, so opposite $= \sqrt{1 - x^2}$.
- So $\sin(\cos^{-1} x) = \sqrt{1 - x^2}$.

5. General Techniques

- Always look for ways to rewrite angles as sums/differences of special angles (like 30° , 45° , 60°).
- For composite angles, use corresponding formulas to expand.

4. Practice Problems with Answers

1. Evaluate $\sin(75^\circ)$ exactly.

Solution: $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$
 $= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4}$

2. Rewrite $5 \sin x + 12 \cos x$ as a single sine function.

Solution: $k = \sqrt{5^2 + 12^2} = 13$ $\sin \phi = \frac{12}{13}$, $\cos \phi = \frac{5}{13}$ So,
 $5 \sin x + 12 \cos x = 13 \sin(x + \phi)$, with $\phi = \arcsin(12/13)$.

3. Prove $\cos(x - y) = \cos x \cos y + \sin x \sin y$ **using the subtraction formula.**

Solution: By definition, the subtraction formula for cosine is:

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

So the identity is verified.

4. Find the exact value of $\tan(75^\circ)$.

Solution: $\tan(75^\circ) = \tan(45^\circ + 30^\circ) = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$
 $= \frac{1 + \frac{1}{\sqrt{3}}}{1 - (1) \cdot \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}+1}{\sqrt{3}}}{\frac{\sqrt{3}-1}{\sqrt{3}}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ Multiply numerator and denominator by $\sqrt{3} + 1$ to rationalize: $= \frac{(\sqrt{3}+1)^2}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{3+2\sqrt{3}+1}{3-1} = \frac{4+2\sqrt{3}}{2} = 2 + \sqrt{3}$

5. If $\sin \theta = \frac{3}{5}$, θ **in Quadrant II, and** $\cos \phi = -\frac{4}{5}$, ϕ **in Quadrant III, find** $\sin(\theta + \phi)$.

Solution:

- In Quadrant II, $\cos \theta = -\sqrt{1 - (\frac{3}{5})^2} = -\frac{4}{5}$.
- In Quadrant III, $\sin \phi = -\sqrt{1 - (-\frac{4}{5})^2} = -\frac{3}{5}$.

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi = \frac{3}{5} \cdot -\frac{4}{5} + (-\frac{4}{5})(-\frac{3}{5}) = -\frac{12}{25} + \frac{12}{25} =$$

5. Assumptions and Ambiguities Resolved

- In a few spots, symbols like ϕ are defined as phase angles in the formula $A \sin x + B \cos x$, which is clarified here.

- The identities for sine, cosine, and tangent additions are standard and have been confirmed.
 - Where the document had formatting or missing explicit equalities (likely was a PDF OCR issue), standard trigonometric identities and formulas have been used.
 - For Quadrant signs, standard ASTC (All Students Take Calculus; meaning All positive in I, Sine in II, Tangent in III, Cosine in IV) principles were applied for the sine/cosine evaluation.
 - Problems and answers created strictly use the information from the lesson outline.
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End of Study Guide